

Ch 3 Derivatives

In the beginning of Chapter 2, we discussed how to determine the slope of a curve at a point and how to measure the rate at which a function changes. Now that we have studied limits, we can define these ideas precisely and see that both are interpretations of the derivative of a function at a point. We then extend this concept from a single point to the derivative function, and we develop rules for finding this derivative function easily, without having to calculate any limits directly.

3.1 Tangents and the Derivative at a Point

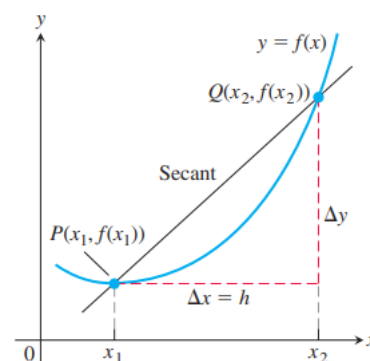
In this section we define the slope and tangent to a curve at a point, and the derivative of a function at a point. The derivative gives a way to find both the slope of a graph and the instantaneous rate of change of a function.

① Instantaneous Rates of Change of a Function

The average rate of change of $y = f(x)$ with respect to x over the interval $[x_1, x_2]$ is

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Geometrically, the rate of change of f over $[x_1, x_2]$ is the slope of the secant PQ .



Let's consider what happens as the point Q approaches the point P along the curve.

Assume $h = \Delta x = x_2 - x_1$, then $Q \rightarrow P \Leftrightarrow h \rightarrow 0$. Using the definition of limit, we get

an expression

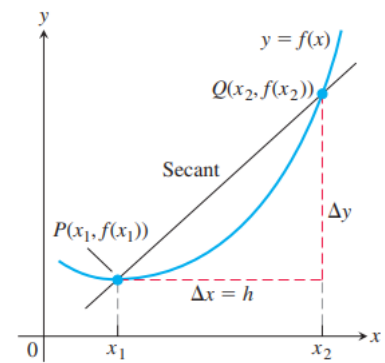
$$\lim_{h \rightarrow 0} \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}.$$

If the above limit exists, it is called **instantaneous rates of change at P** .

② Tangent Line to the Graph of a Function

To find a tangent line to an arbitrary curve $y = f(x)$ at a point $P(x_0, f(x_0))$, we need to find the slope of the tangent line first. To do it, we use the procedure introduced in Section 2.1.

We calculate the slope of the secant line through P and a nearby point $Q(x_0 + h, f(x_0 + h))$. We then investigate the limit of the slope as $h \rightarrow 0$. If the limit exists, we call it the slope of the curve at P (the slope of the tangent at P) and define the tangent line at P to be the line through P having this slope.



DEFINITION 1 The slope of the curve $y = f(x)$ at the point $P(x_0, f(x_0))$ is the number

$$m = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided the limit exists.

The **tangent line to the curve at P** is the line through P with this slope. That is, the

tangent line to the curve at P is

$$y - f(x_0) = m(x - x_0).$$

Example 1

- (a) Find the slope of the curve $y = \frac{1}{x}$ at any point $x = a \neq 0$. What is the slope at the point $x = -1$?
- (b) Where does the slope equal $-\frac{1}{4}$?
- (c) What happens to the tangent to the curve at the point $(a, \frac{1}{a})$ as a changes?

Example 2 Let $f(x) = x^2$. Find the tangent line of $f(x)$ at $x = 1$.

③ Derivative at a Point

Both ① and ② lead to the limit:

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

The expression

$$\frac{f(x_0 + h) - f(x_0)}{h}, h \neq 0$$

is called the difference quotient of f at x_0 with increment h . If the difference quotient has a limit as h approaches zero, that limit is given a special name and notation.

DEFINITION 2 The derivative of a function f at a point x_0 , denoted $f'(x_0)$, is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

provided this limit exists.

If we interpret the difference quotient as the slope of a secant line, then the derivative gives **the slope of the curve** $y = f(x)$ at the point $P(x_0, f(x_0))$.

If we interpret the difference quotient as an average rate of change, the derivative gives the function's **instantaneous rate of change** with respect to x at the point $x = x_0$.

Example 3 Using the definition to find the derivative of $y = -2x^2$ at $x = 1$ and find the equation of the tangent line at the point $(1, -2)$.

Summary

We have been discussing slopes of curves, lines tangent to a curve, the rate of change of a function, and the derivative of a function at a point. All of these ideas refer to the same limit.

We know that the slope of a curve at a point P on the curve is the slope of the tangent. The following are all interpretations for the limit of the difference quotient,

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h} .$$

1. The slope of the graph of $y = f(x)$ at $x = x_0$.

2. The slope of the tangent to the curve $y = f(x)$ at $x = x_0$.
3. The instantaneous rate of change of $f(x)$ with respect to x at $x = x_0$.
4. The derivative $f'(x_0)$ at a point.